

Course Code: CS1081		
Name of the Programme: BCA (Hons.)		
Name of the Course: Programming with C		
Semester: I		
Course Credit	No. of Hours Per Week (L + T + P)	Total no. of Teaching Hours
4	3:0:2	75
Course Objectives:		
a) Introduce the fundamental principles of programming and develop essential problem-solving skills.		
b) Build proficiency in core C programming constructs, including control structures, functions, arrays, pointers, and strings.		
c) Enable students to write, debug, and optimize efficient C programs through practical implementation.		
d) Lay a strong foundation for exploring advanced computing topics and further programming languages.		

Course Outline (Syllabus of the Course)

Module - 1 (Introduction to Programming and C Language Basics)	No. of Hours
	9 Hrs.
Introduction to Programming, History and features of the C language, Structure of a C program, Compilation and Execution of a C program, Basic syntax: keywords, identifiers, constants, variables, and data types, Operators and expressions, Input and output functions: printf (), scanf().	
Module - 2 (Control Structures Topics)	No. of Hours
	9 Hrs.
Decision-making statements: if, if-else, nested if-else, switch, Looping statements: while, do-while, for, break, continue and go to statements.	
Module - 3 (Functions and Arrays)	No. of Hours
	9 Hrs.
Functions: declaration, definition, and calling, Function types: with/without arguments, with/without return value, Storage classes, Recursion, Arrays: one-dimensional and two-dimensional arrays.	
Module - 4 (Pointers and Strings)	No. of Hours
	9 Hrs.
Introduction to pointers, Pointer arithmetic, Pointers and arrays, Pointers to functions, Strings: declaration, initialization, and string handling functions.	

Module - 5 (Structures, Unions, and File Handling)	No. of Hours
	9 Hrs.
Structures: definition, declaration, and initialization, Arrays of structures, Pointers to structures, Unions, File handling: file operations, reading from and writing to files.	

Course Outcomes:
a) Understand the fundamental concepts and basic syntax rules of C programming.
b) Analyze decision-making and looping constructs, and apply string manipulation techniques to solve programming problems.
c) Apply functions, arrays, and pointers, including dynamic memory allocation, to develop modular and efficient code.
d) Create user-defined data types such as structures and unions to organize and manage complex data effectively.

Text Book
1 Balagurusamy, E. (2019). <i>Programming in ANSI C</i> (8th ed.). McGraw Hill Education. ISBN: 978-9351343202
2 Kanetkar, Y. (2022). <i>Let us C</i> (19th ed.). BPB Publications. ISBN: 978-8183331630

Reference Books
1 Kernighan, B. W., & Ritchie, D. M. (2015). <i>The C programming language</i> (2nd ed.). Prentice Hall. ISBN: 978-9332549449
2 Schildt, H. (2017). <i>C: The complete reference</i> (4th ed.). McGraw Hill Education. ISBN: 978-0070411838

Laboratory Component	30 Hrs.
1. Write a C program to print your name entered by the user.	
2. Write a C program to perform arithmetic operations.	
3. Write a C program to swap 2 numbers a. without using a temporary variable b. using a temporary variable	
4. Write a C program to find the largest of three numbers.	
5. Write a C program to print the first 10 Fibonacci numbers.	
6. Write a C program to print the following Half Pyramid of Numbers pattern. 1 1 2 1 2 3 1 2 3 4 1 2 3 4 5	
7. Write a C program to calculate the factorial of a number.	
8. Write a C program to check whether a number is prime.	
9. Write a recursive function to find the GCD of two numbers.	
10. Write a function to perform matrix addition and multiplication.	

11.	Write a program to swap two numbers using pointers.
12.	Write a program to reverse a string.
13.	Write a C program to Find the Frequency of Characters in a String.
14.	Write a program to store and display information of a student using structures.
15.	Write a program to read and write a file.
16.	Write a program to count the number of lines, words, and characters in a file.